

Virtual Reality Based Mass Disaster Triage Training for Emergency Medical Services

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ABSTRACT

When mass disasters with multiple casualties and injured people happen, the Emergency Medical Service staff needs to have strong organizational skills in addition to medical knowledge in order to efficiently apply triage systems. Currently, it is very expensive for Emergency Medical Services such as the Red Cross to practice these skills, because they need to set up elaborate training scenarios with actors and complex environments. In addition, only one person can train at a time, after which the initial scenario must be restored. As a result, only very few people can profit from the training.

This work presents a Virtual Reality training application that aims to replicate the learning experience of the real-life training without the disadvantage of the organizational effort. The application is meant to complement the current training allowing for a more frequent training of mass disaster emergencies for all staff members.

Index Terms: Training, teaching, entertainment—User experience and usability—3D user interaction—;

1 INTRODUCTION

Although the historical development of Virtual Reality (= VR) goes back to the 1960s, advances in digital technologies brought a big advance to the capabilities of VR applications and equipment in recent years [8] [6]. Inexpensive systems are finding their way into different areas, including immersive environments for educational and training tools. New potential for cost reduction, efficiency or risk reduction, as well diverse application possibilities unfold in various industries: above all in the industrial production, healthcare, construction and the creative industries.

VR training systems offer several advantages. VR systems can be used to acquire new abilities or improve existing ones. The use of VR technologies for training has been found to provide a range of benefits compared to traditional training systems. It allows teaching under the learning-by-doing approach when constraints such as e.g. safety, time, or costs prevent the use of real environments [7] [9]. In addition, VR systems can provide extra clues, not available in the real world, that can facilitate the learning of the task (e.g. by giving additional visual, auditory, or haptic feedback); and they allow simulating the task in a flexible way to adapt it to users' needs and training goals [7] [9]. VR systems are used to provide practical training in a number of human activities, including medical training [3] [7] [9], rehabilitation [2], flight [4], or industrial design [10].

This paper presents a VR training environment to practice the necessary skills for Emergency Medical Service first responders in case of a multi casualty incident. The goal of this project was to create a VR application in which the same issues could be trained compared to the elaborate real-life training, but without the disadvantage of the organizational effort and scarce availability. The application is



Figure 1: The virtual scenario as seen from above.

meant to complement the current training in that it allows for a more frequent training of these rescue operations. The concept of the virtual scenario was created by the Red Cross and features a multi casualty incident on a highway (see Figure 1). With the application, trainees can practice using the triage tag to categorize the health status of patients, gain hazard-awareness and practice organizing the accident site.

2 CURRENT RED CROSS MULTI CASUALTY INCIDENT TRAINING

When an accident happens, the emergency vehicle that is closest is sent to the accident site. In case of a mass casualty, a head of operations is necessary to keep an overview and organize medical staff. Until the arrival of the head of operations, the first responders from Emergency Medical Services such as the Red Cross must take over the role of the head of operations. The tasks of the head of operations include organizing the accident site, evaluating the condition of involved people based on a predefined set of medical parameters using a triage tag and report back to the control centre. Heads of operations receive a special education and training to manage these kinds of accidents. Regular Red Cross staff members do not receive that kind of training but still have to take on the role of a temporary head of operations until the real one arrives on site.

Current training scenarios to practice the skills of the temporary head of operations are highly resource-intensive. They require an elaborate set-up of fake accident sites that include a lot of people acting as though they were in an accident. Because of the resource-intensive nature, only few staff members can complete this training.

2.1 The Triage Tag

The Red Cross uses triage tags as guidance to categorize people that have been involved in an accident. The term “triage tag” is used to describe cards that are hung around the patient’s neck in the event of a mass casualty incident for clear identification and to support care. This is a Red Cross standard and must be used by the entire rescue force. The patients are treated by following rescue forces according to their assigned category which is displayed on the triage tag:

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Figure 2: The triage tag (left) and its implementation as 3D User Interface in VR (right). Image has been altered for translation as app was created in German.

- *Green* means they suffer from only small wounds (or none) and are able to walk
- *Orange* means patients suffer from a bleeding wound that is stoppable
- *Red* means they don't have sufficient breathing or suffer from an unstoppable bleeding and therefore need urgent help

2.2 Bringing Triage Training to VR

In order to say whether VR is a suitable platform for Red Cross training scenarios, the application needed to be as similar to the real-life training as possible. The training scenario was designed by the Red Cross and represents a multi casualty accident on the highway. The scenario consists of multiple crashed cars and 10 people that have been involved in the accident (see Figure 1). The scenario also features more or less prominent objects that could be dangerous for following rescue forces arriving later.

Just like during the real-life training, trainees can interact with patients and need to categorize them by using a triage tag (see Figure 2). Visual and auditory cues are given so that trainees can categorise them based upon their behaviour, look and statements. Visual clues include animations like holding the injured body part, waving, etc., but also texturing of the 3D-Character models and the surrounding environment (wounds, blood stains).

The training simulation that was created is meant to be used by two people and can be described as asymmetrical gameplay [1]. One staff member of the Red Cross (= "the operator") who already knows the simulation takes on the role of the control centre. Operators are responsible for starting the scenario and helping trainees in case of questions regarding the simulation or the VR headset. During the training, the part of the operator is to demand status updates or answers radio calls from the trainee. The trainee is the one who wears the VR headset and moves and interacts within the virtual environment.

Trainees have 8 minutes to complete the training scenario. That is the average time it takes for the designated head of operations to arrive at real accident sites. After using the application, the trainee receives a debriefing. This debriefing is crucial for the learning effect as talking about what the trainee experienced and receiving feedback helps memorizing important aspects of the training.

2.3 Virtual Reality Interaction

The application was created in Unity¹ for the HTC Vive². To implement the locomotion and interaction VRTK³ was used. Unique

¹<https://unity.com/>

²<https://www.vive.com/de/product/#vive-series>

³<https://vrtoolkit.readme.io/>

interactions that VRTK did not cover (such as the interaction with the triage tag) were programmed manually.

Staff members of the Red Cross reflect all ages, genders, different backgrounds, etc. It was therefore important to implement interactions that are intuitive or easy to learn for a broad range of people.

To facilitate learning how to use the controls, as few different buttons as possible were used. Meaning that the trigger button provokes various interactions, based upon what the user hovers with a ray-casting pointer [5]. If the user hovers over a patient and presses the trigger button, the patient's monologue can be heard. If the user hovers a potential hazard or the triage tag, a UI becomes visible.

Every patient has its own triage tag that is located close to the patient. Upon categorizing a patient, the square with the corresponding color is displayed above the patient's head so that trainees can keep an overview about the accident site more easily. This also simulates the checklist, that is used for keeping an overview during real-life training or operations.

The application has been tested with 14 students and 27 members of the Red Cross, among them were paramedics and practitioners. Overall, the application has been received very positively. Subjects clearly see a positive learning effect when using the application. It was also shown that the application cannot replace the real-life training but can be a suitable addition for training certain procedures such as using the triage tag. The need for more training of infrequent procedures has been confirmed as subjects could be observed having difficulties understanding and using the triage system correctly.

3 FUTURE WORK

In order to test the concept itself one scenario was deemed to be sufficient, but in future, the application will be extended either by creating multiple different scenarios or by creating a scenario that is adaptable by the operator. With the adaptable scenario, operators could change the number of people involved in the accident, their health status, their location, the number of cars and potential hazards etc. When presenting our work and future ideas to the Red Cross, they expressed their eagerness to continue working on the project.

REFERENCES

- [1] Asymmetric Multiplayer. Library Catalog: tvtropes.org.
- [2] A. Alamri, M. Eid, R. Iglesias, S. Shirmohammadi, and A. El Saddik. Haptic virtual rehabilitation exercises for poststroke diagnosis. *IEEE transactions on instrumentation and measurement*, 57(9):1876–1884, 2008.
- [3] C. Basdogan, S. De, J. Kim, M. Muniyandi, H. Kim, and M. A. Srinivasan. Haptics in minimally invasive surgical simulation and training. *IEEE computer graphics and applications*, 24(2):56–64, 2004.
- [4] M. Blake. The nasa advanced concepts flight simulator-a unique transport aircraft research environment. In *Flight Simulation Technologies Conference*, p. 3518, 1996.
- [5] D. A. Bowman. 3D User Interfaces. In *The Encyclopedia of Human-Computer Interaction*, 2nd Ed., chap. 32.4.4 and 32.5.2. 2013.
- [6] J. Bucher. *Storytelling for virtual reality: Methods and principles for crafting immersive narratives*. Taylor & Francis, 2017.
- [7] A. Derossis, J. Bothwell, H. Sigman, and G. Fried. The effect of practice on performance in a laparoscopic simulator. *Surgical endoscopy*, 12(9):1117–1120, 1998.
- [8] R. Dörner, W. Broll, P. Grimm, and B. Jung. Virtual und augmented reality. *Grundlagen und Methoden der Virtuellen und Augmentierten Realität*. Berlin und Heidelberg: Springer, 2013.
- [9] F. Gosselin, F. Ferlay, S. Bouchigny, C. Mégard, and F. Taha. Design of a multimodal vr platform for the training of surgery skills. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*, pp. 109–116. Springer, 2010.
- [10] O. Portillo-Rodriguez, C. Avizzano, A. Chavez-Aguilar, M. Raspolli, S. Marcheschi, and M. Bergamasco. Haptic desktop: The virtual assistant designer. In *2nd IEEE/ASME International Conference on Mechatronics and Embedded Systems and Applications*. IEEE, 2006.